The Backyard Coop *Ventilation Checklist*

Minimum vent areas, placement rules, warning signs, ammonia thresholds

print it, laminate it, hang it in the coop

MINIMUM VENT AREA BY COOP SIZE

Sized Vent Cheat Sheet

COOP SIZE	BIRDS (EST.)	WINTER MIN	SUMMER MIN	HOT CLIMATE
4 x 4 ft	3 - 4	1.6 sq ft	3.2 sq ft	4.8 sq ft
4 x 8 ft	6 - 8	3.2 sq ft	6.4 sq ft	9.6 sq ft
6 x 8 ft	9 - 12	4.8 sq ft	9.6 sq ft	14.4 sq ft
8 x 10 ft	14 - 20	8.0 sq ft	16.0 sq ft	24.0 sq ft
My coop: ___x___ ft	*___ birds*	_____	_____	_____

Formula: 1 sq ft per bird (winter), 2 sq ft per bird (summer), 3 sq ft per bird (hot climate, >90F avg high). Source: UKy ID-204, Penn State Extension.

PLACEMENT RULES

Where Vents Go

HIGH VENTS **Open eave vents and ridge vents stay open year-round.** Hot, moist air rises and exits here. Closing them traps ammonia and moisture even in winter.

LOW VENTS **Side vents below roost height are "draft vents."** Close or baffle these in winter to prevent cold air hitting roosting birds directly.

EAVE GAPS **Minimum 1-inch gap under eaves on opposite walls** creates cross-flow. Cover with 1/2-inch hardware cloth, not solid boards.

RIDGE VENTS **Best passive option on a gable roof.** 1 linear foot of ridge vent per 4 birds in warm climates, 1 per 6 birds cold climates.

AMMONIA SMELL THRESHOLDS

The Smell Test

None **Good.** Continue current vent setup. Check monthly.

Faint You can smell it. **Ammonia is already above 5 ppm.** Open all high vents immediately. Scrape dropping boards today. Add more ventilation before bedtime.

Strong Eyes watering. Birds are past 25 ppm. **Emergency: add mechanical exhaust fan within 24 hours.** Remove wet bedding now. Call vet if respiratory symptoms.

Key rule: if you smell ammonia at chicken-head height (crouch down), you are already past the threshold. Birds have no way to leave.

FOUR WARNING SIGNS OF BAD VENTILATION

Early Warning Checklist

Frost on inside walls
Moisture from breath and droppings condensing. Too little air exchange. Open high vents wider.

Condensation on combs
Humidity above 70% at roost level. Direct draft risk, but worse: frostbite follows. Check high-vent clearance.

Lethargy in birds
Ammonia above 10 ppm suppresses appetite and immune response. Note if it worsens after cleaning vs. before.

Respiratory rattles
Wheezing, gurgling, or clicking at rest. Can be viral, but ventilation is the first variable to fix. Vet visit warranted.

- ☐ Check inside walls for frost or condensation every winter morning
- ☐ Listen for labored breathing at roost time once per week
- ☐ Test ammonia level monthly (sniff at knee height, or use a test strip)
- ☐ Note litter dryness: wet litter = ventilation or drainage problem